

Spectral Methods for Linear PDEs on a Periodic Domain

Overview

We develop Fourier spectral solvers on a periodic domain $-L < x < L$ for:

- ▶ Advection: $u_t = cu_x$
- ▶ Diffusion: $u_t = ku_{xx}$
- ▶ Advection–Diffusion: $u_t = ku_{xx} + cu_x$

Key ideas:

- ▶ Fourier series representation
- ▶ Diagonalization of spatial derivatives
- ▶ Exact time evolution in Fourier space

Periodic Fourier Representation

Assume $u(x, t)$ is periodic on $[-L, L]$.

$$u(x, t) = \sum_{k=-N/2}^{N/2-1} \hat{u}_k(t) e^{ik\pi x/L}$$

Define wave numbers:

$$\kappa_k = \frac{k\pi}{L}$$

Then:

$$u_x \leftrightarrow i\kappa_k \hat{u}_k \quad \text{and} \quad u_{xx} \leftrightarrow -\kappa_k^2 \hat{u}_k$$

Discrete Fourier Transform

Grid points:

$$x_j = -L + j\Delta x, \quad j = 0, \dots, N-1$$

- ▶ Use FFT to move between physical and spectral space
- ▶ Derivatives computed via multiplication in Fourier space

Advection: $u_t = cu_x$

Fourier transform gives:

$$\frac{d}{dt}\hat{u}_k = ic\kappa_k\hat{u}_k$$

Solution:

$$\hat{u}_k(t) = \hat{u}_k(0)e^{ic\kappa_k t}$$

Advection Solver

Algorithm for one time step Δt :

1. FFT: $u \rightarrow \hat{u}$
2. Multiply: $\hat{u}_k \leftarrow \hat{u}_k e^{i c \kappa_k \Delta t}$
3. IFFT: $\hat{u} \rightarrow u$

Advection Behavior

- ▶ Pure translation at speed c
- ▶ No change in amplitude or shape
- ▶ Spectral method is exact in time for this linear PDE

Diffusion: $u_t = ku_{xx}$

Fourier transform gives:

$$\frac{d}{dt}\hat{u}_k = -k\kappa_k^2\hat{u}_k$$

Solution:

$$\hat{u}_k(t) = \hat{u}_k(0)e^{-k\kappa_k^2 t}$$

Diffusion Solver

Algorithm for one time step:

1. FFT
2. Multiply: $\hat{u}_k \leftarrow \hat{u}_k e^{-k\kappa_k^2 \Delta t}$
3. IFFT

Diffusion Behavior

- ▶ High frequencies decay rapidly
- ▶ Solution smooths over time
- ▶ Eventually approaches spatial mean

Advection–Diffusion: $u_t = ku_{xx} + cu_x$

Fourier transform gives:

$$\frac{d}{dt}\hat{u}_k = (-k\kappa_k^2 + ic\kappa_k)\hat{u}_k$$

Solution:

$$\hat{u}_k(t) = \hat{u}_k(0)e^{(-k\kappa_k^2 + ic\kappa_k)t}$$

Advection–Diffusion Solver

One step:

1. FFT
2. Multiply: $\hat{u}_k \leftarrow \hat{u}_k e^{(-k\kappa_k^2 + ic\kappa_k)\Delta t}$
3. IFFT

Behavior

- ▶ Combination of translation and smoothing
- ▶ Wave travels at speed c
- ▶ Amplitude decays due to diffusion

Eigenvalues λ_k

- ▶ Advection: $\lambda_k = ic\kappa_k$
- ▶ Diffusion: $\lambda_k = -k\kappa_k^2$
- ▶ Advection–Diffusion: $\lambda_k = -k\kappa_k^2 + ic\kappa_k$

Method of Lines Idea

Instead of diagonalizing in Fourier space, we discretize space first:

- ▶ Replace spatial derivatives with finite differences
- ▶ Obtain a system of ODEs in time
- ▶ Integrate using standard ODE solvers

Finite Difference Approximations

Let $x_j = -L + j\Delta x$.

- First derivative (central):

$$u_x(x_j) \approx \frac{u_{j+1} - u_{j-1}}{2\Delta x}$$

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- First derivative (upwind, $c > 0$):

$$u_x(x_j) \approx \frac{u_j - u_{j-1}}{\Delta x}$$

- Second derivative:

$$u_{xx}(x_j) \approx \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$$

Semi-Discrete Systems

This leads to ODE systems:

- ▶ Advection:

$$\frac{du_j}{dt} = cD_x u_j$$

- ▶ Diffusion:

$$\frac{du_j}{dt} = kD_{xx} u_j$$

- ▶ Advection–Diffusion:

$$\frac{du_j}{dt} = kD_{xx} u_j + cD_x u_j$$

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Time Stepping Choices

Common ODE solvers:

- ▶ Explicit Euler (simple, unstable for stiff problems)
- ▶ RK4 (popular, good accuracy)
- ▶ Implicit methods (important for diffusion)

Stability constraints:

- ▶ Advection: CFL condition $c \frac{\Delta t}{\Delta x} \leq C$
- ▶ Diffusion: $k \frac{\Delta t}{\Delta x^2} \leq C$

Implementation Details

Key steps:

- ▶ Store solution as vector u_j
- ▶ Apply periodic boundary conditions:
 - ▶ $u_{-1} = u_{N-1}$
 - ▶ $u_N = u_0$
- ▶ Construct difference operators using indexing

Accuracy and Stability

- ▶ Spectral accuracy in space
- ▶ Exact time stepping for linear constant-coefficient PDEs
- ▶ No CFL restriction for these exact updates

Aliasing and Practical Issues

- ▶ Linear problems: no aliasing issues
- ▶ FFT efficiency: $O(N \log N)$
- ▶ Periodicity is essential

Summary

- ▶ Fourier methods diagonalize linear PDEs
- ▶ Each Fourier mode evolves independently
- ▶ Provides clean insight into physics:
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 - ▶ Advection = phase shift
 - ▶ Diffusion = amplitude decay
 - ▶ Combined = both”